



Lesson 7.12 — Unit 7 Review and Assessment

Big idea: Unit 7 is trig: angles in radians, the unit circle, the three main functions, their graphs, identities, and modeling periodic real-world phenomena.

Quick review

Unit circle: point $(\cos \theta, \sin \theta)$. Memorize Q1 special angles.

Conversions: $\pi = 180^\circ$. Multiply or divide by $\pi/180$.

Graphs: $y = a \sin(b(x - h)) + k$ — amplitude $|a|$, period $2\pi/|b|$, phase shift h , midline $y = k$.

Identities: $\sin^2 \theta + \cos^2 \theta = 1$. Reciprocal: $\csc$, $\sec$, $\cot$.

Inverse trig: principal ranges. Solving full equations may need additional quadrant work.

Worked example

Worked review. Solve $2 \cos \theta = 1$ for $0 \leq \theta < 2\pi$. $\cos \theta = 1/2 \Rightarrow \theta = \pi/3, 5\pi/3$.

Warm-ups

1. Convert 60° to radians.
2. Find $\sin(\pi/4)$.
3. State the period of $y = \sin(2x)$.

Practice

1. Find $\tan(2\pi/3)$.
2. State amplitude, period, midline of $y = 4 \sin(\pi x) - 2$.
3. Solve $\sin \theta = -\sqrt{3}/2$ for $0 \leq \theta < 2\pi$.

Challenge

1. If $\cos \theta = -5/13$ and θ is in Q3, find $\sin \theta$ and $\tan \theta$ exactly.
2. Daylight in a town swings from 10 to 14 hours over a year, with maximum around June 21. Write a sinusoidal model $D(t)$ where t is the day of the year (1–365).



Answer key — Lesson 7.12

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $\pi/3$.
2. $\sqrt{2}/2$.
3. π .

Practice

1. Q2: $\sin = \sqrt{3}/2$, $\cos = -1/2$. $\tan = -\sqrt{3}$.
2. Amplitude 4, period $2\pi/\pi = 2$, midline $y = -2$.
3. Reference $\pi/3$. Q3: $4\pi/3$, Q4: $5\pi/3$.

Challenge

1. $\sin^2\theta = 1 - 25/169 = 144/169 \Rightarrow \sin\theta = \pm 12/13$. Q3: negative $\Rightarrow \sin\theta = -12/13$. $\tan\theta = (-12/13)/(-5/13) = 12/5$.
2. Amplitude 2, midline 12. Period 365. $b = 2\pi/365$. Max at $t \approx 172$ (June 21). $D(t) = 2 \cos((2\pi/365)(t - 172)) + 12$.